

```

CREATE DATABASE ETL_Assignment;
USE ETL_Assignment;
CREATE TABLE Sales_Transactions (
    Txn_ID INT,
    Customer_ID VARCHAR(10),
    Customer_Name VARCHAR(50),
    Product_ID VARCHAR(10),
    Quantity INT,
    Txn_Amount DECIMAL(10,2),
    Txn_Date DATE,
    City VARCHAR(50)
);
CREATE TABLE Customers_Master (
    CustomerID VARCHAR(10),
    CustomerName VARCHAR(50),
    City VARCHAR(50)
);
INSERT INTO Customers_Master VALUES
('C101', 'Rahul Mehta', 'Mumbai'),
('C102', 'Anjali Rao', 'Bengaluru'),
('C103', 'Suresh Iyer', 'Chennai'),
('C104', 'Neha Singh', 'Delhi');
INSERT INTO Sales_Transactions VALUES
(201, 'C101', 'Rahul Mehta', 'P11', 2, 4000, '2025-12-01', 'Mumbai'),
(202, 'C102', 'Anjali Rao', 'P12', 1, 1500, '2025-12-01', 'Bengaluru'),
(203, 'C101', 'Rahul Mehta', 'P11', 2, 4000, '2025-12-01', 'Mumbai'),
(204, 'C103', 'Suresh Iyer', 'P13', 3, 6000, '2025-12-02', 'Chennai'),
(205, 'C104', 'Neha Singh', 'P14', NULL, 2500, '2025-12-02', 'Delhi'),
(206, 'C105', 'N/A', 'P15', 1, NULL, '2025-12-03', 'Pune'),
(207, 'C106', 'Amit Verma', 'P16', 1, 1800, NULL, 'Pune'),
(208, 'C101', 'Rahul Mehta', 'P11', 2, 4000, '2025-12-01', 'Mumbai');

SELECT * FROM Sales_Transactions;
SELECT * FROM Customers_Master;

```

question -7

– Duplicate Business Key

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SELECT
    Customer_ID,
    Product_ID,
    Txn_Date,
    Txn_Amount,
    COUNT(*) AS duplicate_count
FROM Sales_Transactions

```

```
GROUP BY
    Customer_ID,
    Product_ID,
    Txn_Date,
    Txn_Amount
HAVING COUNT(*) > 1;
Q8 – Referential Integrity Violation
```

```
SELECT DISTINCT st.Customer_ID
FROM Sales_Transactions st
LEFT JOIN Customers_Master cm
    ON st.Customer_ID = cm.CustomerID
WHERE cm.CustomerID IS NULL;
SELECT *
FROM Sales_Transactions
WHERE Quantity IS NULL
    OR Txn_Amount IS NULL
    OR Txn_Date IS NULL;
```

Question 1:

Define Data Quality in ETL. Why is it more than just cleaning?

Data Quality in ETL refers to how accurate, complete, consistent, valid, unique, and timely the data is as it flows through Extract → Transform → Load.

It is more than data cleaning because:

Cleaning = fixing errors (nulls, typos, duplicates)

Data Quality = preventing + measuring + governing + validating

It includes:

Validation rules

Monitoring metrics

Integrity checks

Standardization

Business rule enforcement

👉 Cleaning is reactive; Data Quality is proactive + continuous.

Question 2: Why poor data quality leads to misleading dashboards?

Poor data quality causes:

❌ Wrong aggregations

❌ Incorrect KPIs

❌ Misleading trends

✗ False insights

Example:

If duplicates exist → Revenue doubles → Business thinks sales increased.

Outcome:

- ➡ Wrong decisions
- ➡ Financial losses
- ➡ Strategy failures

Garbage In → Garbage Out (GIGO)

Question 3: What is duplicate data? Three causes in ETL

Duplicate Data = Same record appearing multiple times.

Causes:

- 1 Multiple Source Loads
Same file/API loaded repeatedly.
- 2 Missing Business Key Constraints
No uniqueness checks.
- 3 Improper Joins / Transformations
Cartesian joins creating duplicates.

Other causes:

Retry failures

Late arriving data

CDC misconfigurations

Question 4: Exact vs Partial vs Fuzzy duplicates

Type	Meaning	Example
Exact	Fully identical rows	Same Customer_ID, Product_ID, Amount
Partial	Some fields differ	Same ID but different City
Fuzzy	Similar but not identical	"Rahul Mehta" vs "R. Mehta"

Question 5: Why validate during transformation?

Validation during Transformation is better because:

- ✓ Errors caught early
- ✓ Prevents corrupt warehouse
- ✓ Saves reprocessing cost
- ✓ Easier correction

If done after loading:

- ✗ Bad data already impacts reports
- ✗ Costly rollback

👉 Best Practice = Validate before Load

Question 6: Business Rules & Example

Business Rules = Logical constraints based on domain knowledge.

Example Rules:

Txn_Amount > 0

Quantity NOT NULL

Txn_Date NOT NULL

Example from dataset:

C105 → Txn_Amount = NULL ✗

C104 → Quantity = NULL ✗

C106 → Txn_Date = NULL ✗

Business rules help detect:

Invalid values

Missing fields